

AIR LENS

IEEE 802.11 Capability Intelligence & Negotiation Analysis Framework

Air Lens is an IEEE 802.11 Wi-Fi Intelligence Engine designed to analyze wireless packet captures (PCAPs) and extract low-level protocol capabilities directly from over-the-air management frames.

Key Features

- High-speed RawPcapReader filtering
- HE (Wi-Fi 6) and EHT (Wi-Fi 7) capability extraction
- PMF and NSS analysis
- Multi-Link Operation (MLO) decoding
- Three-way capability reconciliation matrix
- Negotiation bottleneck identification

Pipeline Architecture

Raw Wi-Fi PCAP → engine.py → Management Frames → Profilers → ie_parser.py → capability_diff.py → report_formatter.py

Core Modules

engine.py – Fast low-level packet filtering engine
ie_parser.py – Bitwise TLV parser
ap_profile.py – AP capability compiler
client_profile.py – Client capability compiler
assoc_response_parser.py – Association response parser
capability_diff.py – Capability reconciliation engine
report_formatter.py – Matrix visualization framework
run_profiler.py – Main execution coordinator

Installation

```
git clone https://github.com/arunvishal2712/Air-Lens---Blue-Print-tool-for-pcap.git
```

```
python3 -m venv venv  
pip install --upgrade pip  
pip install scapy
```

Conclusion

Air Lens provides a lightweight and extensible framework for deep IEEE 802.11 management-frame intelligence and capability negotiation analysis.